

Exercise Sheet 6 - Solutions

Propositional Logic – SAT

1. Let $A = ((p \wedge q) \rightarrow r) \rightarrow (p \vee \neg r)$. To convert A to a DNF we first compute A 's truth table:

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$	$\neg r$	$p \vee \neg r$	A
T	T	T	T	T	F	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	F	T	T
T	F	F	F	T	T	T	T
F	T	T	F	T	F	F	F
F	T	F	F	T	T	T	T
F	F	T	F	T	F	F	F
F	F	F	F	T	T	T	T

We enumerate all the **T** rows:

- $p \wedge q \wedge r$
- $p \wedge q \wedge \neg r$
- $p \wedge \neg q \wedge r$
- $p \wedge \neg q \wedge \neg r$
- $\neg p \wedge q \wedge \neg r$
- $\neg p \wedge \neg q \wedge \neg r$

Finally, we combine those using $\vee$:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

2. To convert A to a CNF we first enumerate all the **F** rows:

- $\neg p \wedge q \wedge r$
- $\neg p \wedge \neg q \wedge r$

We combine the negation of those using $\wedge$:

$$\neg(\neg p \wedge q \wedge r) \wedge \neg(\neg p \wedge \neg q \wedge r)$$

We then use de Morgan and double negation elimination to convert this formula to a CNF as follows:

- $\neg(\neg p \wedge q \wedge r) \wedge \neg(\neg p \wedge \neg q \wedge r)$
- $(\neg\neg p \vee \neg q \vee \neg r) \wedge \neg(\neg p \wedge \neg q \wedge r)$ – by de Morgan
- $(\neg\neg p \vee \neg q \vee \neg r) \wedge (\neg\neg p \vee \neg\neg q \vee \neg r)$ – by de Morgan
- $(p \vee \neg q \vee \neg r) \wedge (\neg\neg p \vee \neg\neg q \vee \neg r)$ – by DNE

- $(p \vee \neg q \vee \neg r) \wedge (p \vee \neg \neg q \vee \neg r)$ – by DNE
- $(p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r)$ – by DNE

3. We will now convert A to a CNF using equivalences

- $((p \wedge q) \rightarrow r) \rightarrow (p \vee \neg r)$
- $(\neg(p \wedge q) \vee r) \rightarrow (p \vee \neg r)$ – implication elimination
- $\neg(\neg(p \wedge q) \vee r) \vee (p \vee \neg r)$ – implication elimination
- $(\neg\neg(p \wedge q) \wedge \neg r) \vee (p \vee \neg r)$ – de Morgan
- $((p \wedge q) \wedge \neg r) \vee (p \vee \neg r)$ – DNE
- $(p \vee \neg r) \vee ((p \wedge q) \wedge \neg r)$ – commutativity of $\vee$
- $((p \vee \neg r) \vee (p \wedge q)) \wedge ((p \vee \neg r) \vee \neg r)$ – distributivity of $\vee$ over $\wedge$
- $((p \vee \neg r) \vee p) \wedge ((p \vee \neg r) \vee q) \wedge ((p \vee \neg r) \vee \neg r)$ – distributivity of $\vee$ over $\wedge$
- $((p \vee \neg r) \vee p) \wedge ((p \vee \neg r) \vee q) \wedge (p \vee \neg r \vee \neg r)$ – associativity of $\vee$
- $((p \vee \neg r \vee p) \wedge ((p \vee \neg r) \vee q)) \wedge (p \vee \neg r \vee \neg r)$ – associativity of $\vee$
- $((p \vee \neg r \vee p) \wedge (p \vee \neg r \vee q)) \wedge (p \vee \neg r \vee \neg r)$ – associativity of $\vee$
- $(p \vee \neg r \vee p) \wedge (p \vee \neg r \vee q) \wedge (p \vee \neg r \vee \neg r)$ – associativity of $\wedge$

Through idempotence, commutativity and associativity, this could even be simplified further to:

$$(p \vee \neg r) \wedge (p \vee q \vee \neg r)$$

4. Here is a possible run of the algorithm:

- $(p \vee t \vee s) \wedge (q \vee r \vee \neg s \vee \neg t) \wedge (p \vee \neg q \vee s) \wedge (p \vee q \vee r \vee \neg t)$
 $\wedge (q \vee r \vee \neg s) \wedge (\neg p \vee \neg s \vee \neg t) \wedge (\neg p \vee \neg q \vee s \vee \neg r) \wedge (\neg r \vee t)$
 - **select** $p = \mathbf{T}$
 - $(q \vee r \vee \neg s \vee \neg t) \wedge (q \vee r \vee \neg s) \wedge (\neg s \vee \neg t) \wedge (\neg q \vee s \vee \neg r) \wedge (\neg r \vee t)$
 - **select** $q = \mathbf{T}$
 - $(\neg s \vee \neg t) \wedge (s \vee \neg r) \wedge (\neg r \vee t)$
 - **select** $r = \mathbf{T}$
 - $(\neg s \vee \neg t) \wedge (s) \wedge (t)$
 - **select** $s = \mathbf{T}$
 - $(\neg t) \wedge (t)$
 - **select** $t = \mathbf{F}$
 - $()$ – empty clause – **backtrack**
 - **select** $t = \mathbf{T}$
 - $()$ – empty clause – **backtrack**
 - **select** $s = \mathbf{F}$
 - $() \wedge (t)$ – empty clause – **backtrack**
 - **select** $r = \mathbf{F}$
 - $(\neg s \vee \neg t)$
 - **select** $s = \mathbf{F}$
 - no more clauses – **SAT**

Therefore this formula is satisfiable. It is for example satisfied by the valuation:

$$p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{F}, s = \mathbf{F}, t = \mathbf{T}$$